

Importance of Cryptography in Secure Communication

Objective

The objective of this practical is to investigate the transmission of dummy login credentials over HTTP and HTTPS using Wireshark. The captured network packets are analysed to understand how credentials appear in each protocol and to study the importance of encryption in protecting sensitive information.

Procedure

Wireshark was used to capture network traffic through the `eth0` interface. A login request was generated and the captured packets were examined using suitable display filters. The HTTP POST request was analysed to observe whether the submitted login information could be read from the packet. Similarly, HTTPS traffic was inspected to determine whether the application data was visible or encrypted. The observations from both protocols were then compared.

1. HTTP Traffic Analysis

Observation:

The Wireshark capture shows an HTTP POST request from the local system `192.168.254.132` to the destination `44.238.29.244`. The request is transmitted using HTTP over port 80.

When the packet is expanded, the **HTML Form URL Encoded** section displays the submitted form fields. The username and password are visible directly as part of the HTTP request. The packet's raw data section also contains readable characters corresponding to the submitted information.

Explanation:

HTTP does not encrypt the application data transmitted between the client and server. Therefore, when login credentials are submitted through an HTTP connection, they can appear as readable information in a captured packet. Anyone capable of monitoring the network traffic could potentially obtain the credentials.

This demonstrates that HTTP is not suitable for transmitting sensitive information because the communication does not provide confidentiality.

2. HTTPS Traffic Analysis

Observation:

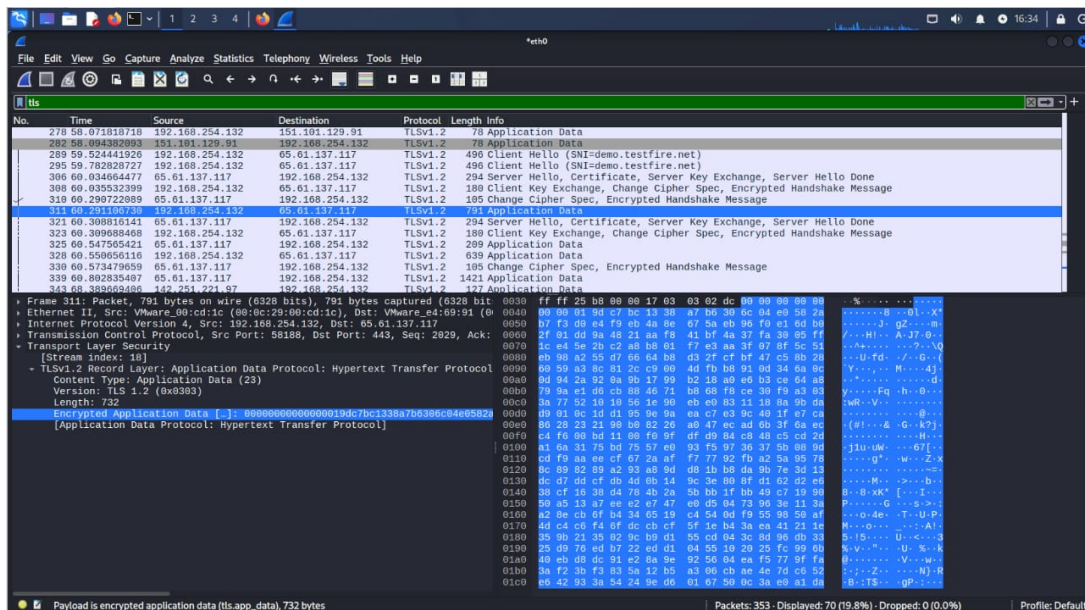


Figure 2: HTTPS TLS Application Data captured using Wireshark

3. Comparison

Aspect	HTTP	HTTPS
Port	80	443
Security	Unencrypted	TLS protected
Credential visibility	Readable in packet	Not directly readable
Data transmission	Plain application data	Encrypted application data
Confidentiality	Not provided	Provided through TLS
Protection of credentials	Low	High

4. Role of Encryption

The experiment demonstrates the importance of encryption in network communication. In the HTTP capture, the login information could be observed directly because the data was transmitted without encryption.

In the HTTPS capture, TLS encryption protected the application data from being directly understood. Encryption therefore prevents captured network traffic from easily revealing sensitive information such as usernames and passwords.

HTTPS also provides additional security through data integrity and server authentication, making it more suitable for secure communication.

5. Conclusion

The Wireshark experiment clearly demonstrated the security difference between HTTP and HTTPS. The HTTP POST request exposed the submitted login information in readable form, while the HTTPS capture contained encrypted TLS application data that could not be directly interpreted.

Thus, encryption plays a crucial role in protecting sensitive information during transmission. HTTPS should be used instead of HTTP whenever login credentials or other confidential information are exchanged over a network.